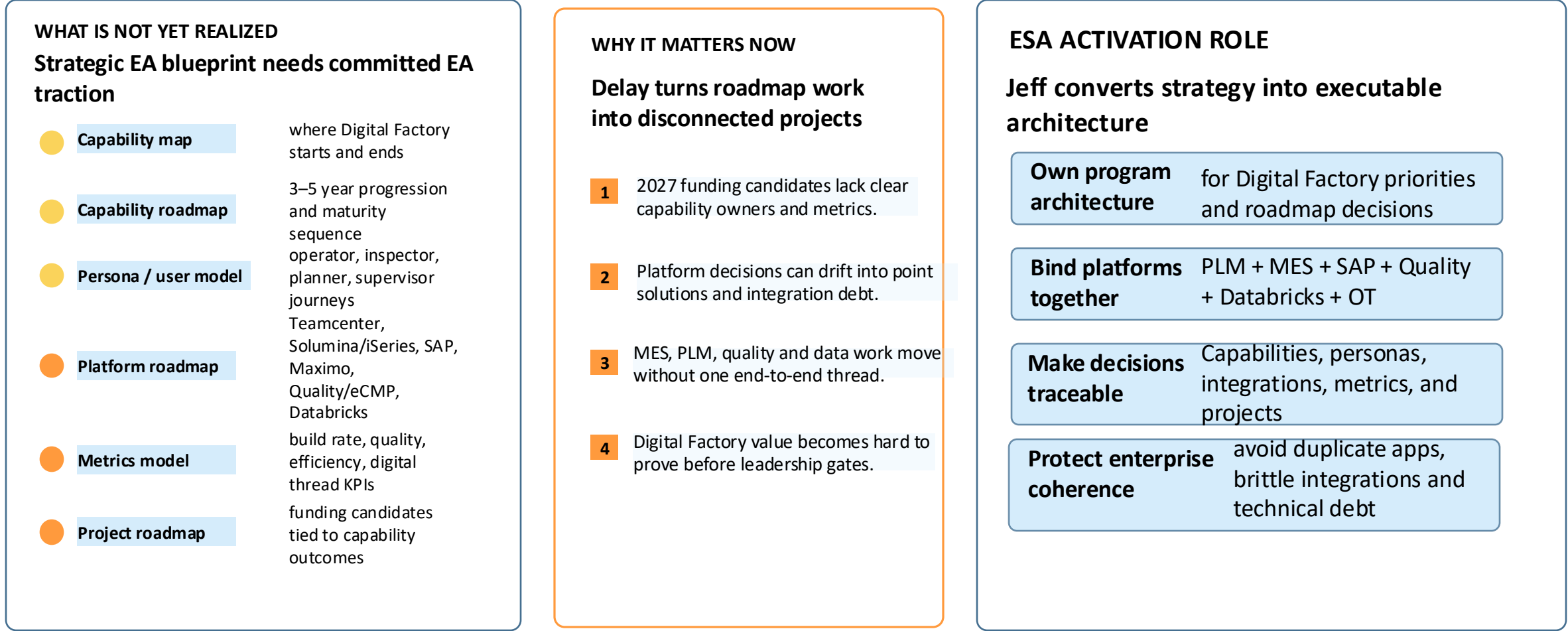


Digital Factory ESA: The Missing Activation Role

Aug./Sept. reality: the strategic EA blueprint exists in pieces — but still needs an accountable owner to convert it into executable architecture.



GULFSTREAM · DIGITAL FACTORY · NEXTGEN OPERATIONS

Jeff Walk: Digital Factory Enterprise Architect for NGO

A rare fit: proven Gulfstream EA experience, manufacturing scale-up, architecture governance, and data/AI architecture depth.

WHY JEFF IS A STRONG FIT

Digital Factory needs an ESA who can operate across shop floor, data, platforms and governance.



Gulfstream-ready EA

Rebuilt EA function in 90 days; established governance charter, standards, ARB, tools and measures.



Manufacturing scale-up

Led regulated manufacturing facility build-out and production scale-up; quality gates and supplier controls.



Governance with pace

Established ARB, review standards and reusable patterns to prevent costly implementation mistakes.



Data + AI depth

Built semantic knowledge layer for 30K+ data assets; graph/RAG experience for governed AI and decision context.

FIRST 90 DAYS: BLUEPRINT TO EXECUTION

Give Jeff the mandate to turn the NGO roadmap into decision-ready architecture artifacts.

0–30 days

Reset blueprint baseline

Validate Digital Factory capabilities, business outcomes and platform boundaries; lock owners and decision gates.

31–60 days

Produce the architecture package

Capability map, persona/user model, integration blueprint, data domain model and platform roadmap.

61–90 days

Tie projects to measurable value

Project-to-capability roadmap, KPI baselines, risk/debt register and leadership decks.

Outcome: one accountable architecture narrative connecting digital thread, MES, quality, data, and shop-floor execution.

GULFSTREAM · DIGITAL FACTORY OF THE FUTURE

Transformation Strategy & Platform Foundation

A solution architecture roadmap unifying design, production, quality, and support under one governed digital thread.

STRATEGY

AI, digital twin, semantic data — with one governed architecture, so every domain from supply chain to MRO draws on the same model and the same twin state.

3 → 1

Foundational pillars unified under one architecture

1 Thread

Per aircraft: configuration → build → completion → service life

100%

Decisions carry lineage — FAA / EASA audit-ready by design

CAPABILITIES — SIX DOMAINS, ONE OPERATING MODEL

Digital Supply Chain

Semantic view of demand, capacity & parts, raw material to delivery.

30–40% inventory cost ↓

Shop Floor & MES

Cells and lines that reason and re-balance in real time.

100% paperless floor

Quality Intelligence

From inspection-based detection to AI prediction & prevention.

25–35% first-time quality ↑

MRO & Predictive Mtn.

Maintenance driven by condition & prediction, not calendar intervals.

40–50% unplanned downtime ↓

Customer Support

From reactive tickets to proactive, twin-informed engagement.

360° unified customer view

On-Demand Repair

Networked repair planned against live aircraft-asset twins.

91% first-time-fix target

PLATFORM BACKBONE — PLM + MANUFACTURING OPERATIONS

Teamcenter — Collaborative PLM (cPLM)

Siemens · Engineering source of truth

- BOM, configuration, requirements & engineering change management for every tail number
- Feeds design intent into the digital thread at “Configured” — traceability begins at Day 0, before a part is cut
- Integrates to shop-floor & MRO systems through the canonical model / event mesh, not point-to-point interfaces

iBase-t Solumina — Cloud-Native MOM Platform

Cloud-native · microservices · built for aerospace & defense

- Unifies MES, embedded Quality (EQMS), Supplier Quality (SQM) and MRO in one operational data model
- Connects engineering intent to shop-floor execution — guides production and enforces quality, not just records it
- Cloud-native, microservices architecture scales across global sites while preserving compliance, security & continuity

MES

EQMS

SQM

MRO

GULFSTREAM · DIGITAL FACTORY OF THE FUTURE

Digital Twin, Digital Factory & Reference Architecture

From component-level twins to an enterprise twin network — grounded in a seven-layer governed architecture.

DIGITAL TWIN — FIVE-LEVEL HIERARCHY

Level	Scope	Update Cadence
Component Twin	Parts & subassemblies	Real-time (ms)
Asset Twin	Equipment, tooling, fixtures	Near real-time (sec)
Process Twin	Operations & workflows	Event-driven (min)
Facility Twin	Plant layout, material flow	Periodic (hrs)
Enterprise Twin	Multi-facility, supply network	Aggregated (daily)

METRICS & KPIS — BASELINE → TARGET (PH.4–5)

KPI	Baseline	Target
First-time quality	base	+32%
Unplanned downtime (idx)	100	40
MRB cycle time (idx)	100	38
Customer response time (idx)	100	52
Decision lineage coverage	10%	100%
AI autonomy phase	P0	P4

DIGITAL FACTORY — SITE NETWORK (TWIN MATURITY)

Savannah, GA	HQ & Final Assembly		52%
Appleton, WI	Completions Center		38%
St. Louis, MO	Service Center (MRO)		34%
Dallas, TX	Service Center (MRO)		30%
Long Beach, CA	Service Center (MRO)		26%
Brunswick, GA	Service Center (MRO)		24%

REFERENCE ARCHITECTURE — 7 LAYERS

L1	Edge & IoT
L2	Integration & Connectivity
L3	Data Platform & Semantic Layer ★
L4	AI / ML Services
L5	Digital Twin
L6	Governance Plane (cross-cutting, all layers)
L7	Applications — PLM · MES · EQMS · SQM · MRO